Transport-scheme comparison — chain vs. SC vs. octree, with and without delta-M

Comparison of the three transport configurations of the `rt3d` solver on the MANUAL §11 worked example, each run with and without delta-M (Wiscombe) scaling. Generated by `scripts/compare_transport.py` (plan: `plan/compare-transport-plan.md`); all figures and CSVs live in `plot/compare_transport/`. Measured 2026-10-13 on macOS/arm64 with the `pyarts` interpreter; iteration counts and physics numbers are machine-independent, wall times are not.

Reproduce with:

```
~/miniforge3/envs/pyarts/bin/python scripts/compare_transport.py
```

(3D HTML scenes additionally need `trame-pyvista`, installed in `pyarts`.)

1. Purpose and run matrix

The solver currently offers three transport algorithms and an optional delta-M split of the scattering operator. This study answers three questions on one physically non-trivial case (strongly forward-scattering cloud, solar beam, reflecting ground):

1. **chain vs. SC** — how large is the pure *transport-algorithm* error when both run on *identical cells*?
2. **unigrid vs. octree** — the grid-refinement error (validation anchor 5).
3. **delta-M on/off** — does it accelerate convergence, and does the converged solution stay the same (the equivalence claim of `plan/delta-m-plan.md`)?

Six runs, one per row:

run key	grid	transport	delta-M
`uni_chain_dM0`	unigrid 24^3	lattice-stencil chains	off
`uni_chain_dM1`	unigrid 24^3	lattice-stencil chains	M = 1
`uni_sc_dM0`	unigrid 24^3	single-cell short characteristics	off
`uni_sc_dM1`	unigrid 24^3	single-cell short characteristics	M = 1
`octree_dM0`	octree (25 264 leaves)	exact characteristic marches	off
`octree_dM1`	octree (25 264 leaves)	exact characteristic marches	M = 1

2. The setups

2.1 Common physical case (MANUAL §11, unchanged)

Gaussian cloud + Lambertian ground + solar beam on the domain x, y ∈ [−5, 5], z ∈ [0, 10] m, periodic in x and y, space at the top, ground at the bottom. Units: lengths m, κ in m^{-1}, radiometric quantities normalised to the incident beam flux $F_0 = 1$.

- **Extinction** κ_e = C·gauss(σ = 1 m, μ = (0, 0, 4) m), C scaled so the maximum vertical column optical depth is τ = 10 (peak κ_e ≈ 3.9 m^{-1}, mean free path ≈ 0.26 m).
- **Scattering** κ_s = ω_local·κ_e with ω_local = gauss/max(gauss), capped at ω ≤ 0.9 (`omega_cap=0.9`) so plain lambda iteration stays tractable.
- **Asymmetry** g_local = gauss/(1.05·max) → peak g ≈ 0.952 in the cloud core (HG phase function) — the showcase regime for delta-M.
- **Thermal emission** B = 0 (pure scattering/beam problem).
- **Solar beam**: direct collimated beam, 30° from zenith, propagating in the x-z plane: ω_sun = [sin 30°, 0, −cos 30°], $F_0 = 1$.
- **Boundary conditions**: isotropic background I = 1 on the top face; Lambertian ground, albedo 0.3, emissivity 0 (no thermal emission).

2.2 Direction grid and iteration scheme (identical in all six runs)

- Icosahedral direction grid, refinement 1 → **42 ordinates** (development grid), solid-angle weights from spherical-triangle areas, Σw = 4π (−7.1e−15).
- Lambda iteration with **1D Ng acceleration**, convergence on the relative source change, tol = 1e−6, cap 60 iterations.

2.3 Grids

- **Unigrid**: 24^3 cells, h = 0.4167 m, 13 824 cells.

- **Octree**: base cell h_0 = 1.25 m; refinement driven by the *physical* κ_e (τ_max = 2.0 per cell, ε_grad = 1.0, max depth 3), 2:1 balanced; **25 264 leaves** (L0: 316, L1: 788, L2: 3 680, L3: 20 480), cell size 1.25 m in the vacuum tails down to 0.156 m inside the cloud.

2.4 Transport algorithms

- `chain` **(unigrid)** — lattice-stencil chain sweeps (`rt3d/characteristics.py`, DOIT-style): per ordinate one integer stencil s, characteristics run center-to-center across the whole grid in multi-cell chains. Exact for piecewise-constant sources along the chain; carries the known *grazing-ordinate stencil error* on coarse grids (MANUAL §6, anchor 5).
- `sc` **(unigrid)** — single-cell short characteristics (`rt3d/short_characteristics.py`, `plan/plan-sc-unigrid.md`): gap segment + target segment of equal length, bilinear 4-point face interpolation, wrap-around passes on periodic entry axes. Local and gather-only; more diffusive on coarse grids (path error ~ linear in cell size).
- **Octree** — exact characteristic marches (`transport.precompute_octree_march` + `sweep_direction_octree_exact`): one backward ray march per leaf per ordinate, no lattice-stencil approximation. Short characteristics are *not available* on the octree (the solver raises `ValueError` for `transport="sc"` there).

2.5 delta-M scaling (`delta_M=1`, delta-Eddington)

Wiscombe split of the HG phase function, defined on the *row-normalized* discrete operator (MANUAL §9, `plan/delta-m-plan.md`):

```
Phi_delta = Phi_HG - (1/4pi) * sum_{n<1} (2n+1) g^n P_n(Omega.Omega')
kappa_e_delta = kappa_e - kappa_s * g^M        (diffuse transport only)
```

with the explicit `-omega·g^M·I/(1 - omega·g^M)` term in the source. Only the scattering part of the extinction is scaled; the direct beam keeps its physical attenuation, and octree refinement stays driven by the physical κ_e. delta-M changes the *iteration* (fewer iterations at g ≈ 0.95), not the converged discrete solution — the on/off comparison below verifies this.

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3. Diagnostics

All cell fields are sampled onto one **common 64³ lattice** at the finest octree cell (h = 0.15625 m) — exact for piecewise-constant fields — so every pair of runs is directly pointwise comparable.

- **Mean intensity** $J = (1/4\pi)\int I\, d\Omega$.
- **Angular variance** $\sigma^2(r) = (1/4\pi)\int (I - J)^2\, d\Omega$ (`rt3d.diagnostics.angular_variance`) — zero for an isotropic field, measures the angular anisotropy of the radiation field.

- **Upward radiance at TOA**: the upward flux map $F^{+}(x, y) = \int_{\mu>0} I\, \mu\, d\Omega$ at the top layer, and the domain-mean upward radiance per upward ordinate (21 of the 42 ordinates have $\mu > 0$).

Six comparison pairs (B − A): SC↔chain per δ, δ-on↔δ-off per method, octree↔unigrid-chain per δ.

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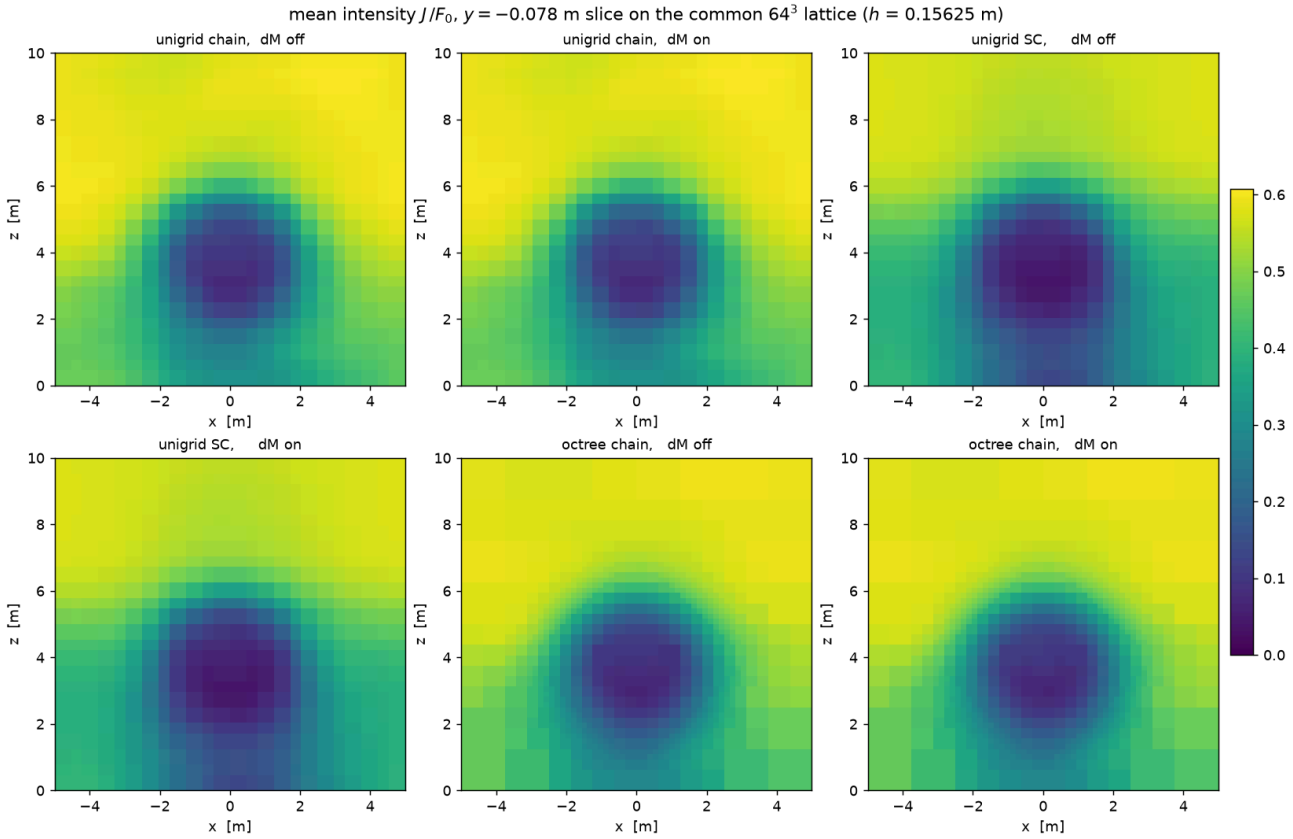
4. Results

4.1 Convergence and efficiency

run	transport	δ-M	n_iter	converged	wall [s]	⟨J⟩	mean F_z @TOA
`uni_chain_dM0`	chain	off	32	✓	1.7	0.5188	−2.5215
`uni_chain_dM1`	chain	M=1	**12**	✓	0.8	0.5188	−2.5214
`uni_sc_dM0`	sc	off	51	✓	18.0	0.4632	−2.5935
`uni_sc_dM1`	sc	M=1	**12**	✓	4.4	0.4633	−2.5933
`octree_dM0`	octree march	off	30	✓	42.9	0.4014	−2.5312
`octree_dM1`	octree march	M=1	**12**	✓	23.9	0.4016	−2.5308

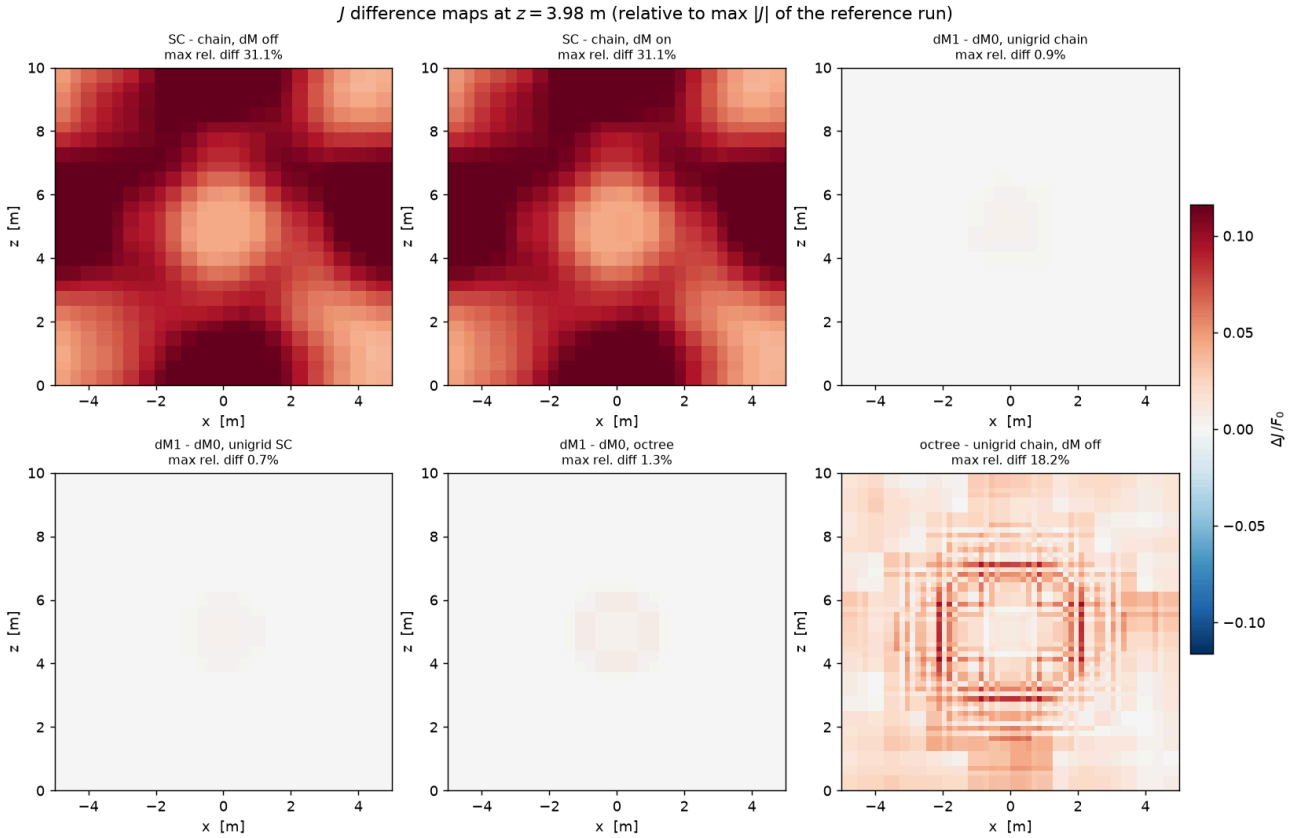
- **delta-M is a large accelerator in this regime**: 32 → 12 (chain), 51 → 12 (SC), 30 → 12 (octree) iterations; wall time drops by ~2× everywhere and by 4× for SC.
- The unigrid TOA mean F_z reproduces the MANUAL §11 numbers exactly (−2.5215 unigrid, −2.5312 octree).
- SC needs notably more iterations than chain *without* delta-M (51 vs 32) and is ~10× slower per iteration on this grid; with delta-M all three schemes converge in 12 iterations.

4.2 Mean intensity J



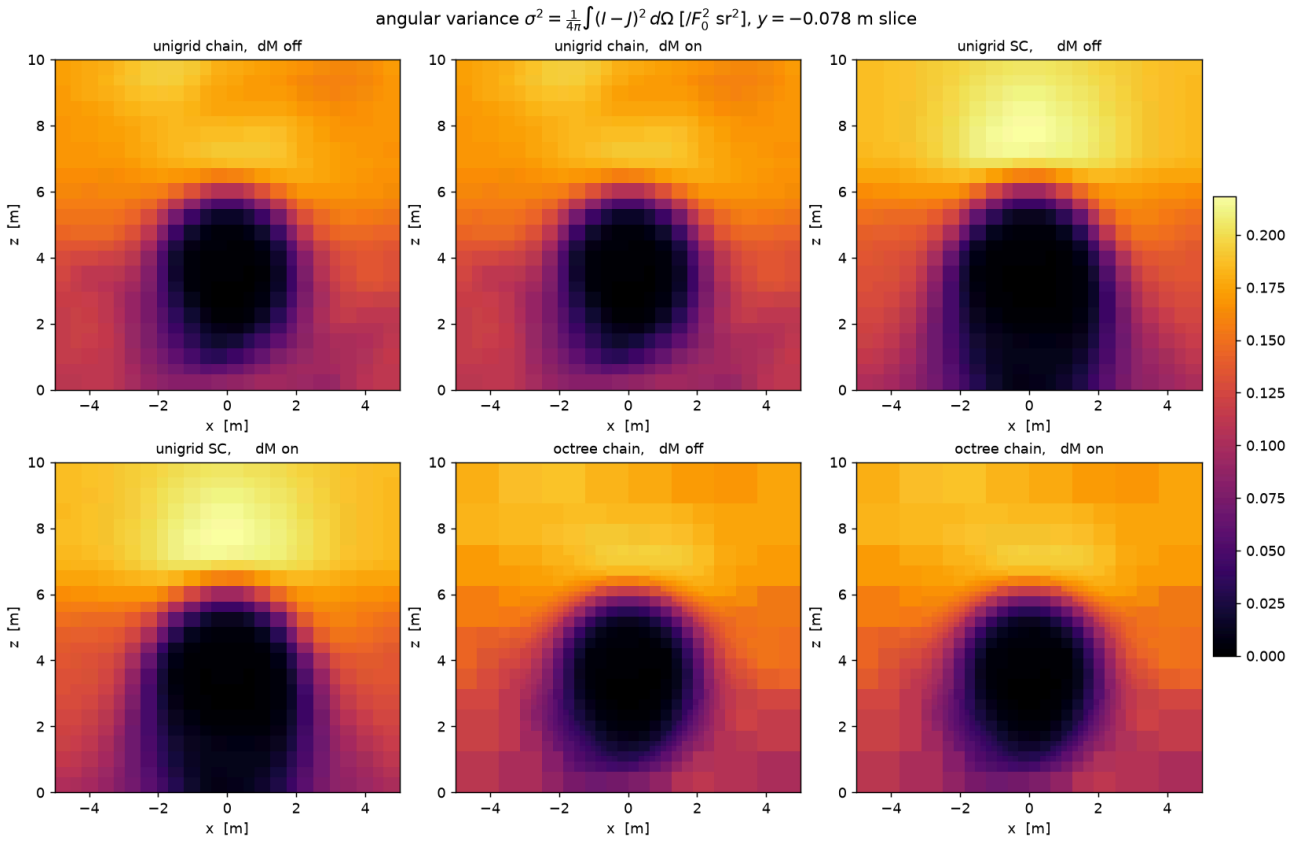
Band-mean J profiles (8 base-cell bands, `profiles.csv`):

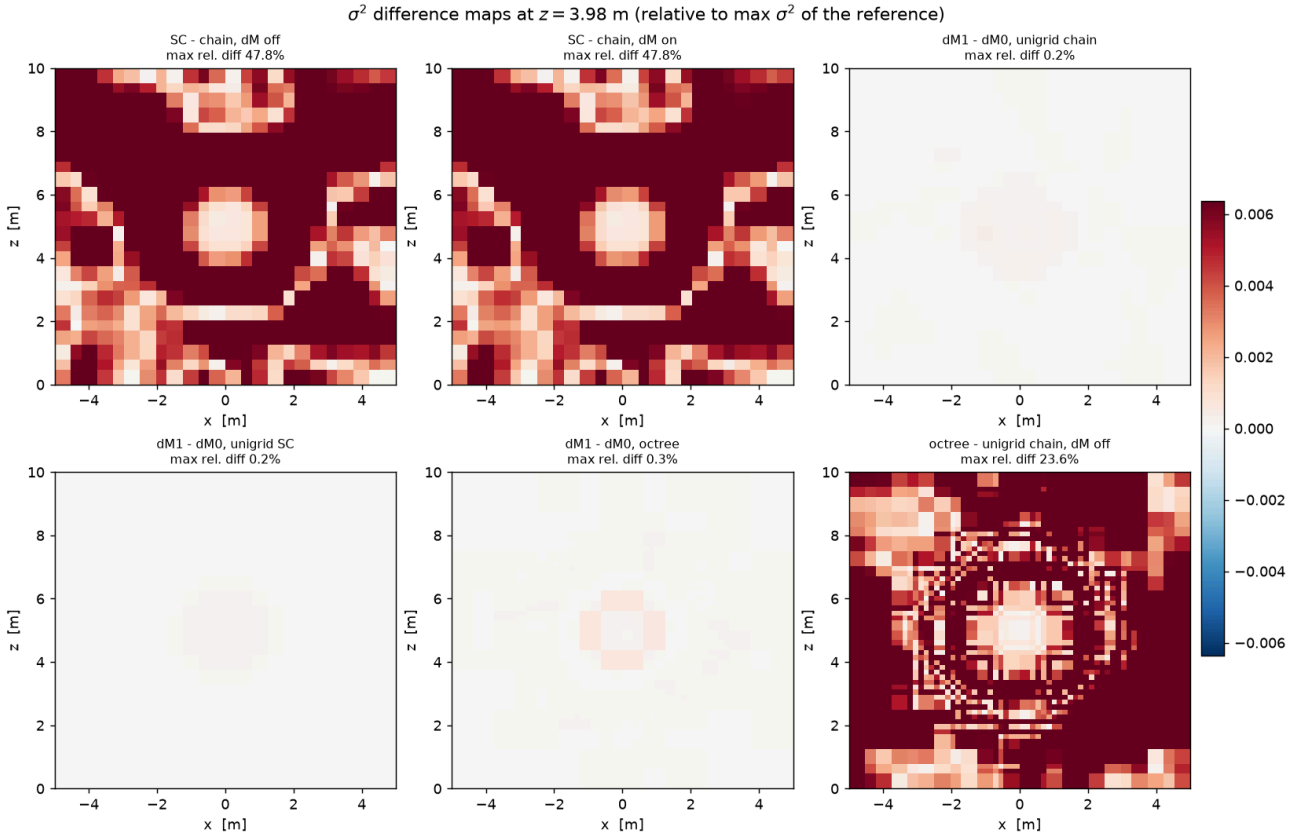
z [m]	chain δ-off	chain δ-on	SC δ-off	SC δ-on	octree δ-off	octree δ-on
0.625	0.4372	0.4372	0.3680	0.3680	0.3095	0.3097
3.125	0.4529	0.4530	0.3629	0.3630	0.3431	0.3434
4.375	0.4964	0.4966	0.4077	0.4078	0.4154	0.4156
6.875	0.5870	0.5870	0.5631	0.5631	0.5377	0.5377
9.375	0.5893	0.5893	0.5712	0.5712	0.5850	0.5851



- **chain vs. SC on identical cells is the largest discrepancy of the study** (max 31%, mean 9%): SC's extra numerical diffusion lets more radiation penetrate the optically thick core, where it is absorbed (ω = 0.9, τ = 10) — J is lower everywhere below and inside the cloud (0.368 vs 0.437 in the bottom band).
- **unigrid-chain vs. octree** (18% max, 2.8% mean) reproduces the anchor-5 pattern of the manual: the unigrid grazing-ordinate stencil error inflates J below the cloud (0.437 vs 0.310 in the bottom band, the manual's 29%), with a ring structure around the cloud in the difference maps.
- **delta-M pairs are flat** (≤1.3% max, ~0.01–0.02% mean): the converged solution is independent of δ-M, as designed; the residual pointwise gap is the documented O(τ_cell) quadrature error of the explicit −ω g^M I term.

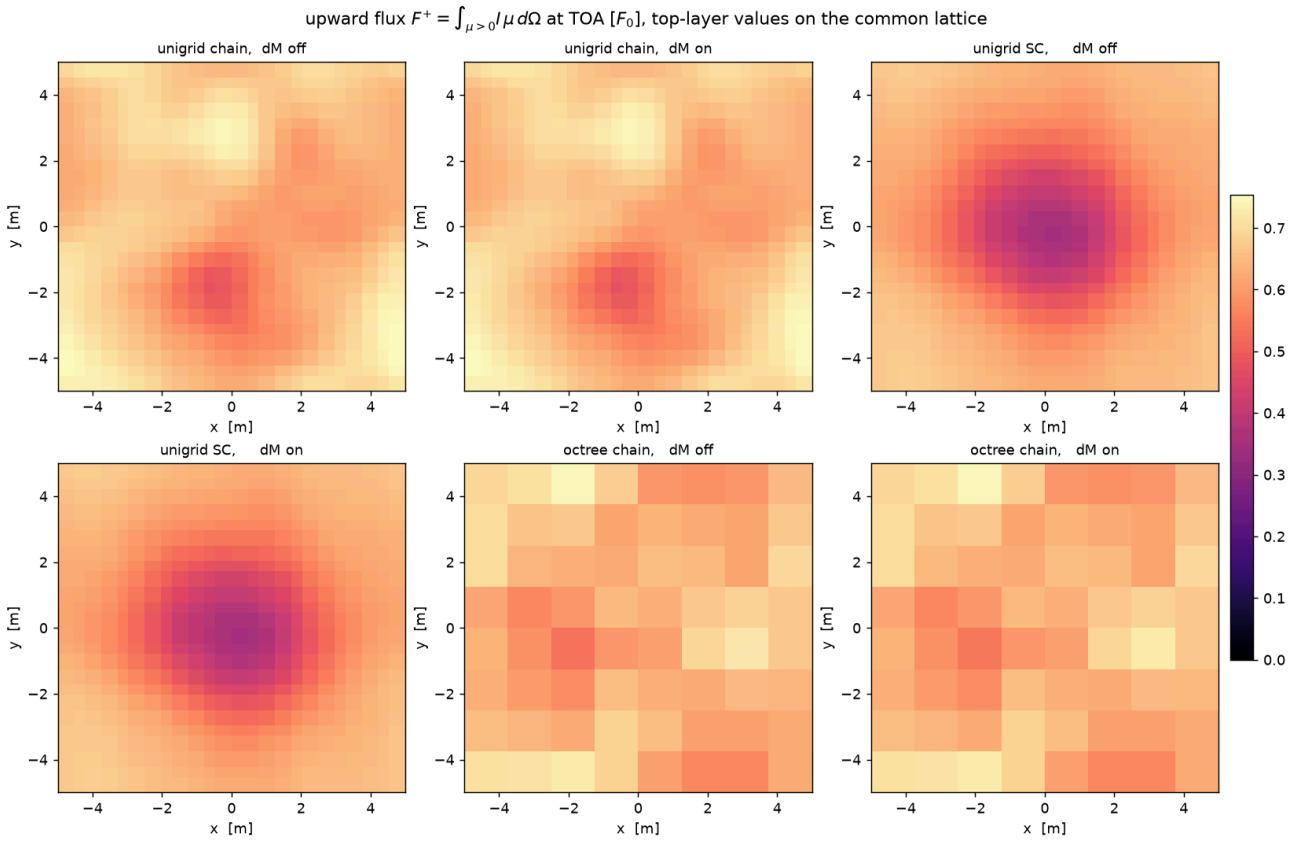
4.3 Angular variance

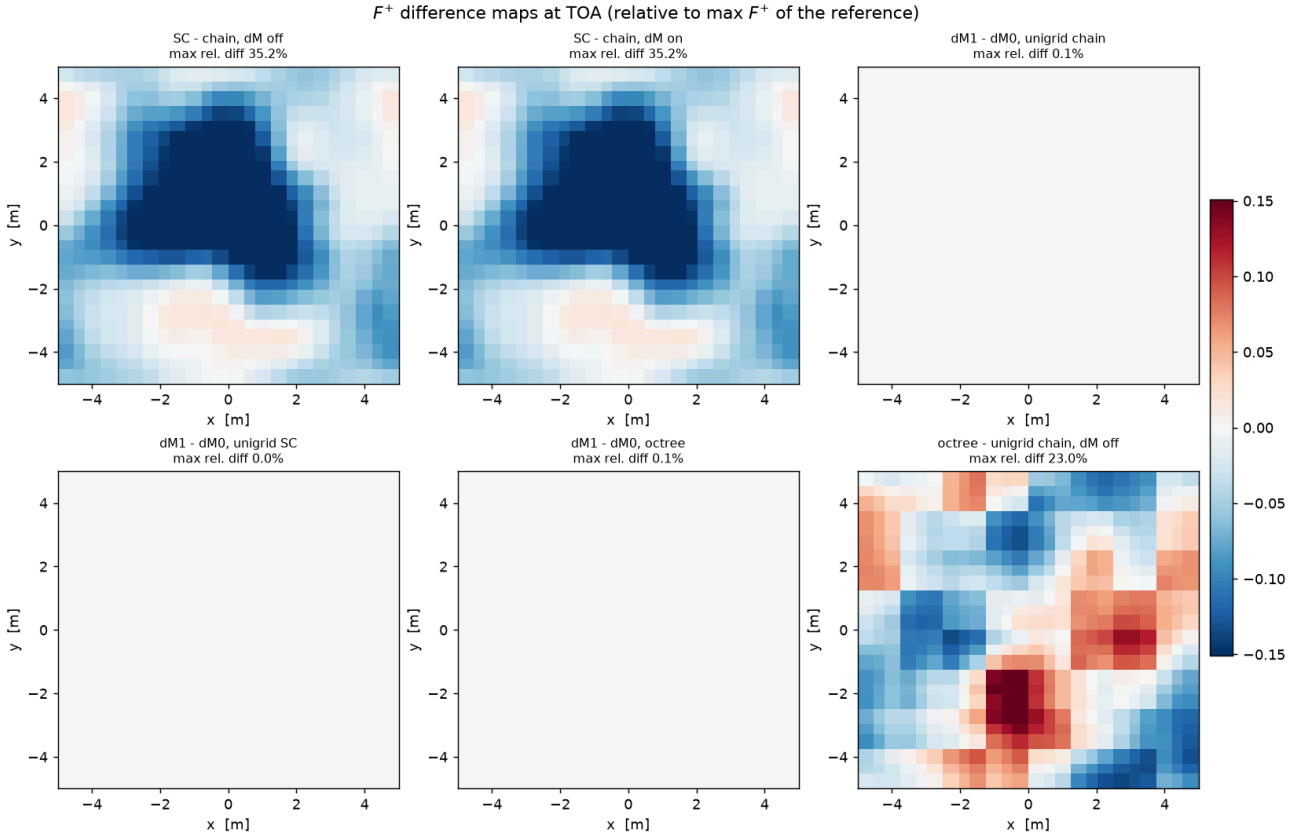


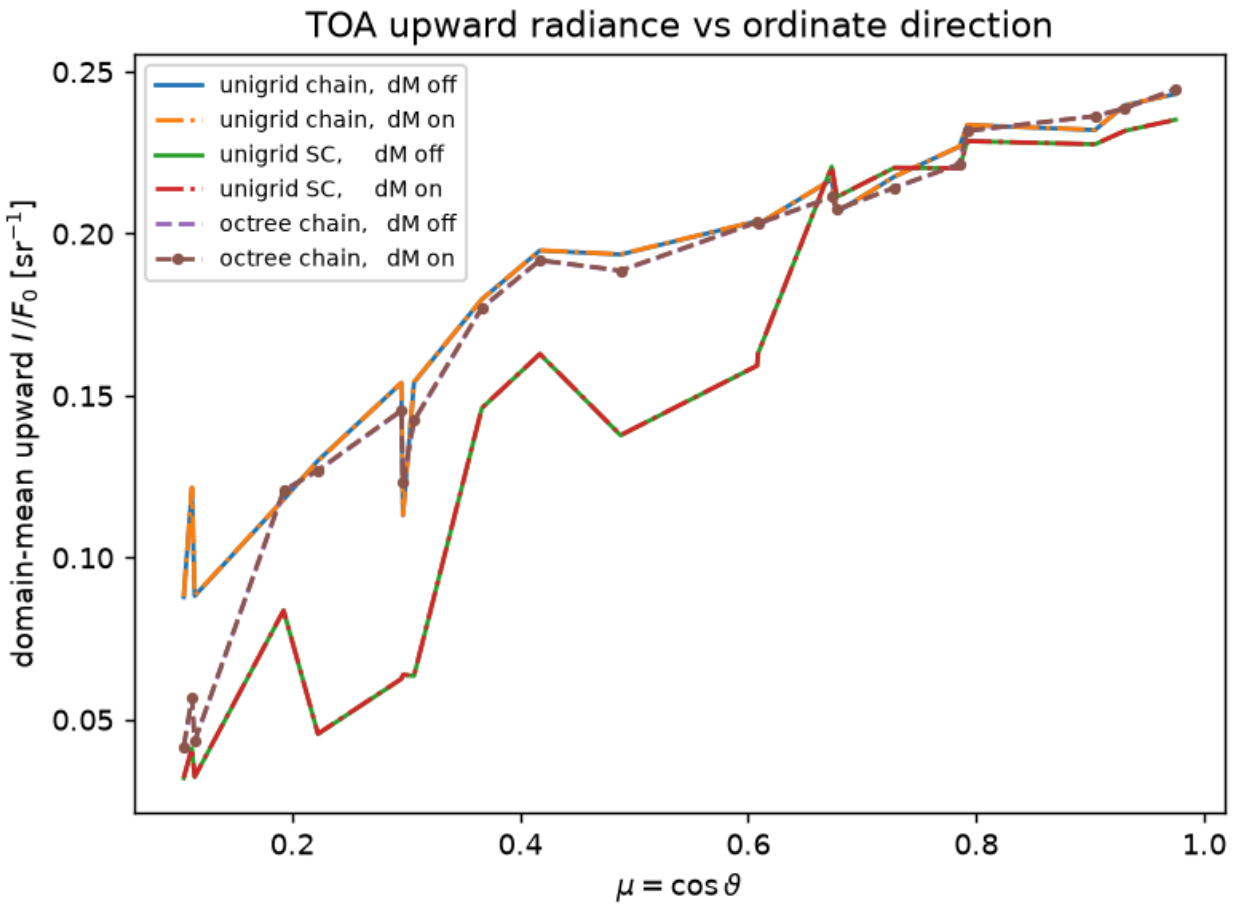


σ^2 is high in the vacuum (collimated beam + background = strongly anisotropic field) and collapses toward zero inside the cloud core where multiple scattering isotropises the field — the "hole" in all six panels. Scheme differences show up mostly in the *shadow region below the cloud* and in the beam path: SC's diffusion smooths the beam shadow (47.8% max variance difference vs. chain), while the octree resolves the shadow boundary at 0.156 m. delta-M changes σ^2 by ≤0.3%.

4.4 Upward radiance at TOA




F^+ difference maps at TOA (relative to max F^+ of the reference)
SC - chain, dM off
max rel. diff 35.2%
SC - chain, dM on
max rel. diff 35.2%
dM1 - dM0, unigrid chain
max rel. diff 0.1%
dM1 - dM0, unigrid SC
max rel. diff 0.0%
dM1 - dM0, octree
max rel. diff 0.1%
octree - unigrid chain, dM off
max rel. diff 23.0%
x [m]
y [m]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15



- The **chain** runs show a *bright* spot above the cloud (the scattering layer seen from above), the **SC** runs a deep *shadow* blob — the TOA-integrated signature of SC's extra diffusion (max F^+ difference 35%). The octree map is coarse by construction (top-layer cells are 1.25–2.5 m, the cloud is far below and the refinement is κ_e-driven).
- Per-ordinate domain-mean radiance (`toa_mean_radiance.csv`): chain and octree agree closely at all μ; SC collapses at grazing ordinates (μ = 0.10: 0.032 vs 0.088 chain / 0.041 octree) — the SC wrap-ring / grazing-path error — and lies between chain and octree at μ ≳ 0.7.
- Domain-mean upward radiance: chain 0.179, octree 0.170, SC 0.142 [F_0 sr^{-1}]; consistently, the TOA net flux is least upward-escaping for SC (mean F_z −2.5935 vs −2.5215/−2.5312).

4.5 Pair statistics (from `pair_stats.csv`)

Relative differences of B − A, normalised by max|A| of the reference run (max over the common lattice / mean over the domain):

pair	J max	J mean	σ^2 max	σ^2 mean	F^+ max	F^+ mean
SC − chain, δ off	31.1%	9.1%	47.8%	6.2%	35.2%	9.8%
SC − chain, δ on	31.1%	9.1%	47.8%	6.2%	35.2%	9.8%
δ1 − δ0, unigrid chain	0.9%	0.01%	0.2%	0.02%	0.1%	0.02%
δ1 − δ0, unigrid SC	0.7%	0.01%	0.2%	0.01%	0.0%	0.02%
δ1 − δ0, octree	1.3%	0.02%	0.3%	0.04%	0.1%	0.05%
octree − unigrid chain, δ off	18.2%	2.8%	23.6%	3.2%	23.0%	7.6%

5. Findings

1. **delta-M (M = 1) does exactly what it should**: it cuts the iteration count to 12 for every scheme (from 32/51/30), and the converged fields agree with the δ-off runs to ≤1.3% max / ≤0.05% mean — the residual is the known O(τ_cell) explicit-term quadrature gap, not a physics change. At the cloud-core g ≈ 0.95 it should be the default setting.
2. **The transport-algorithm error dominates everything else on this 24^3 grid.** chain vs. SC differ by up to 31% in J and 35% in the TOA radiance — larger than the unigrid-vs-octree refinement error (18%). Both errors are first-order-ish in h and angular (grazing ordinates); they are *discretization* differences, not physical ones.
3. **SC is systematically more diffusive**: lower J inside and below the cloud, deeper TOA shadow, collapsed grazing-ordinate radiance, and a ~3% smaller net upward flux at TOA. It is also ~10× slower per iteration on the unigrid (wrap-around ring passes).
4. **The octree mainly buys resolution where it matters** (cloud interior, shadow boundary at 0.156 m) at 1.8× the unigrid cell count; its exact-march transport removes the chain's grazing-ordinate stencil error, which is why unigrid-chain vs. octree is the manual's anchor-5 pattern.
5. **δ-M is scheme- and grid-agnostic**: the δ-on↔δ-off differences are ~2 orders of magnitude smaller than any scheme comparison, so the six-run comparison is not confounded by the scattering-operator split.

6. Interactive 3D scenes

Self-contained vtk.js HTML (open directly in a browser — orbit, actor visibility toggling, colour maps; no server, no Python needed). One scene per run (κ_e isosurfaces at 0.5/1/2/3 m^{-1} + J and σ^2 cut planes at x = 0, y = 0, z = 4) plus two difference scenes (|ΔJ| isosurfaces + slices):

```
plot/compare_transport/scene_uni_chain_dM0.html   scene_uni_chain_dM1.html
plot/compare_transport/scene_uni_sc_dM0.html      scene_uni_sc_dM1.html
plot/compare_transport/scene_octree_dM0.html      scene_octree_dM1.html
plot/compare_transport/scene_diff_chain_vs_sc_dM0.html
plot/compare_transport/scene_diff_uni_vs_octree_dM0.html
```

(pyvista's vtk.js export cannot carry volume mappers, hence slices + isosurfaces; the static volume overview is `vol_j.png`.)

7. Output files

file	content
`summary.csv`	run table (iterations, wall, ⟨J⟩, TOA F_z)
`profiles.csv`	band-mean J profiles, all six runs
`toa_mean_radiance.csv`	per-ordinate domain-mean upward radiance at TOA
`pair_stats.csv`	max/mean relative differences of all six pairs × J, σ^2, F^+
`j_slices.png`, `j_profiles.png`, `j_diff.png`	mean-intensity figures
`var_slices.png`, `var_diff.png`	angular-variance figures
`toa_fup_maps.png`, `toa_fup_diff.png`, `toa_mu_spectra.png`	TOA radiance figures
`vol_j.png`	static volume renderings, all six runs
`scene_*.html`	interactive 3D scenes (gitignored, regenerate by rerunning)